Vishal Yadav

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Professional Summary

Data Science and Analytics fresher with hands-on experience in end-to-end ML pipelines, predictive modeling, EDA, ETL development, and business intelligence reporting. Proficient in Python, SQL, Power BI, and Tableau. Built and tuned classification and regression models achieving **90%+ accuracy**. Passionate about transforming large datasets into actionable business insights through data-driven solutions and compelling visualizations.

Technical Skills

Languages	Python, SQL
ML & Algorithms	Classification, Regression, Clustering (K-Means), Feature Engineering, Hyperparameter Tuning, Cross-Validation, SHAP Interpretability, EDA
ML Libraries	Scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy, Optuna
Visualization & BI	Power BI, Tableau, Matplotlib, Seaborn
Databases	PostgreSQL, MySQL
ETL & Pipelines	Data preprocessing, feature transformation, pipeline automation, model serialization
Tools & Platforms	Excel (Pivot Tables, Power Query, XLOOKUP), Jupyter Notebook, Git, GitHub

Professional Experience

Deloitte — Jul 2025
Data Analytics Virtual Intern — Forage — *Remote*

- Built structured ETL processes in Excel and Tableau to clean, transform, and validate raw business datasets, ensuring data integrity for downstream analytics.
- Analyzed large-scale client operational data to surface time-series trends and anomalies, producing data-driven recommendations that informed business process improvements.
- Designed interactive Tableau dashboards communicating KPIs and performance narratives to stakeholder audiences, improving reporting clarity and decision-making speed.
- Collaborated cross-functionally to align analytical outputs with business requirements and maintain consistent data quality standards across multiple datasets.

Projects

Diabetes Classification | ***Python, Scikit-learn, XGBoost, LightGBM, Optuna*** | ***GitHub*** — *2026*

- Built a full end-to-end ML pipeline on the Pima Indians Diabetes dataset (768 records) covering data ingestion, feature engineering, model training, and evaluation.
- Engineered BMI categories and age bins; applied statistical outlier detection on Insulin and SkinThickness, improving baseline accuracy by **~14 percentage points**.
- Benchmarked 7 algorithms (LR, KNN, CART, RF, GBM, XGBoost, LightGBM); LightGBM + Optuna tuning achieved the top accuracy of **90.07%**.
- Applied SHAP analysis to identify Glucose, Insulin, and BMI as primary predictors; delivered visualizations of distributions, outliers, and class balance.

Customer Segmentation & Value Analysis | ***Python, Pandas, K-Means, Power BI, Seaborn*** | ***GitHub*** — *2025*

- Designed an unsupervised analytics pipeline using K-Means clustering on transactional data to identify customer segments by spending frequency and monetary value.
- Engineered RFM-style features (Recency, Frequency, Monetary) with Pandas to enhance cluster separability and improve business interpretability of segments.
- Isolated high-value cohorts driving the majority of revenue; produced Power BI dashboards to support targeted marketing and customer retention strategies.

AI Impact on Jobs & Layoff Risk Analysis | ***Python, Pandas, Seaborn, Matplotlib, EDA*** | ***GitHub*** *2026*

- Curated and analyzed a dataset of 9,000+ job records across 15+ industries to quantify AI automation exposure scores and layoff risk levels by role, sector, and geography.
- Performed in-depth EDA — uncovering that roles in IT, Finance, and Manufacturing face the highest displacement risk — and visualized trends using Seaborn and Matplotlib for stakeholder-ready reporting.
- Engineered derived features including risk tier classification and industry vulnerability index from raw job market data, enabling multi-dimensional analysis of AI's economic impact.
- Published a structured, reproducible dataset and end-to-end analysis notebook on GitHub/Kaggle, contributing an open resource for workforce strategy and AI policy research.

Certifications

Data Analytics Job Simulation — Deloitte via Forage Jul 2025
Machine Learning for Data Science Projects — IBM SkillsBuild Mar 2026
Data Classification — IBM SkillsBuild Mar 2026
Artificial Intelligence Fundamentals — IBM SkillsBuild Mar 2026
Google Analytics Certification — Google Skillshop Jan 2026

Key Achievements

- Achieved **90.07% accuracy** on the Diabetes Classification pipeline by engineering features and tuning LightGBM with Optuna, a ~14 pp improvement over the baseline.
- Delivered **lowest RMSE and highest R^2** among 7+ regression models in the Insurance Cost project through systematic cross-validated hyperparameter search.
- Completed **4 end-to-end data science projects** across supervised learning, clustering, SQL analytics, and SHAP interpretability, all published on GitHub.
- Earned **4 industry certifications** in ML, AI, and Data Analytics from IBM SkillsBuild and Deloitte, demonstrating consistent upskilling commitment.

Education

Shree L R Tiwari College of Engineering 2023 – 2027
B.Tech in Electronics and Telecommunication Engineering *Mumbai, Maharashtra*